

Project Planning Sheet - Rising Waters

1. Project Milestone Schedule

mermaid

gantt

title Rising Waters Development Timeline

dateFormat YYYY-MM-DD

section Data & Model Phase

Data Collection & Cleaning :active, 2026-06-01, 7d

Exploratory Data Analysis (EDA) : 2026-06-08, 5d

Model Training & Evaluations : 2026-06-13, 6d

section Web App Development

Form Design & Bootstrap 5 Theme : 2026-06-19, 4d

Flask Integration & Routing : 2026-06-23, 4d

Validation Engine & JS overlays : 2026-06-27, 3d

section Testing & Deployment

Unit & Integration Testing : 2026-06-30, 2d

GitHub Setup & Vercel Upload : 2026-07-02, 2d

2. Activity Breakdown

1. Sprint 1: Data Preparation

- Import meteorology records.
- Clean empty values and capping outliers using Interquartile Range (IQR).
- Visualize data correlations with heatmaps in Jupyter Notebook.

2. Sprint 2: Model Engineering & Scaling

- Partition data into 75% training and 25% testing.
- Evaluate Decision Tree, Random Forest, KNN, and XGBoost classifiers.
- Optimize the scaler and export parameters.

3. Sprint 3: Web Server Implementation

- Create Flask app structure with /Predict, /chance, /no_chance routes.

- Structure templates with Bootstrap 5.
- Set up client-side form validations.

4. Sprint 4: Verification & Cloud Launch

- Run local unit tests verifying redirects and validators.
- Refactor codebase to fit Vercel bundle restrictions.
- Initialize git, commit changes, and launch on Vercel.